

Assignment - I
BS in Analytics and Sustainability Studies (2024-28)
Climate Change, Sustainability and Development
Tata Institute of Social Sciences, Mumbai

Assignment Weightage: 50%

Assigned Case Studies

Enrol. No.	Name	Country
M2024BSASS001	Aarushi Roy	United States
M2024BSASS002	Aditi Sharma	China
M2024BSASS003	Aditi Ranjan	India
M2024BSASS004	Anjali Arya	Germany
M2024BSASS005	Arhant Doongarwal	United Kingdom
M2024BSASS007	Avni Sawant	France
M2024BSASS008	Bhogi Yug Hrishikesh	Japan
M2024BSASS009	Boopalan Senthilkumar	Canada
M2024BSASS010	Divy Gauba	Australia
M2024BSASS011	Hriman Saha	Brazil
M2024BSASS012	Jayaditya Sharma	Mexico
M2024BSASS013	Jeel Markana	Pakistan
M2024BSASS014	Keya Sunil Parashar	Indonesia
M2024BSASS015	Krishna Mitra	Nigeria
M2024BSASS016	Lakshita Purwar	Bangladesh
M2024BSASS017	Merry Geordy	South Africa
M2024BSASS018	Nidhi Yadav	Saudi Arabia
M2024BSASS019	Novid Salhot	Iran
M2024BSASS020	Prabudh Singh	Turkey
M2024BSASS022	Ravi Pratap Singh	Vietnam
M2024BSASS023	Riya Rahul Rao	Thailand
M2024BSASS024	Satvik Bhardwaj	Philippines
M2024BSASS025	Sherick	Argentina
M2024BSASS026	Shikhar Srivastava	Chile
M2024BSASS027	Shreyit Sinha	Egypt
M2024BSASS028	Shubhra Bansal	Kenya
M2024BSASS029	Subha Varma	Ethiopia
M2024BSASS030	Utkarsh Rangari	Norway
M2024BSASS031	Vidita Ganesh	Sweden

M2024BSASS032	Wali Abidi	Netherlands
M2024BSASS033	Yeshashwini Kavikondala	South Korea

Each student has been assigned one country. You will collect emissions and development data, analyse trends, visualize patterns, and interpret what they imply about climate responsibility and global inequality.

1. Data Sources – Use credible global datasets such as:

- World Bank Data Portal
- Our World in Data
- Global Carbon Project
- UN Data or UNDP

2. Dataset Requirements

Compile a dataset for your assigned country for 20 years (2005–2025) including:

- Year
- Total CO₂ emissions
- Population
- Per capita emissions
- GDP per capita (recommended)

3. Tasks

a) Country Profile and Data Compilation (10 marks)

Briefly introduce your country and present the dataset you compiled. Include sources and explain the variables used.

b) Emissions Trend Analysis (10 marks)

Create two time-series charts:

- Total CO₂ emissions (2005–2025)
- Per capita CO₂ emissions (2005–2025)

Interpret the trends and discuss possible drivers such as economic growth, industrialization, or policy changes.

c) Carbon Inequality Analysis (15 marks)

Calculate and analyse:

- Per capita emissions of your country
- Share of global emissions
- Comparison with India and global averages

Discuss whether your country is a high, moderate, or low emitter and what this implies for climate responsibility.

d) Visualization and Interpretation (10 marks)

Create a dashboard showing:

- Chart comparing per capita emissions with other countries
- Scatter plot (GDP per capita vs emissions)
- Chart showing emissions share

Explain what these visuals reveal about carbon inequality.

e) Climate Justice Reflection (5 marks)

- Write a short 200–300 word reflection addressing: Based on your analysis, what role should your country play in global climate mitigation efforts? Use data from your analysis to support your argument.

Files need to be submitted on TISSOL:

- Report* - should include:
 - Introduction (Country overview)
 - Dataset and sources
 - Emissions trend analysis
 - Carbon inequality analysis
 - Visualizations and interpretation (Use figures/charts from dashboard)
 - Climate justice reflection
- Dashboard* (as link or excel file)
- Datasets* (as an excel file)

Submission Guidelines

- ▶ Format: Typed document (PDF) and name your file as “*Name_Enrol No._Country name*”
- ▶ Length: 2500 - 3000 words, Font: Times New Roman, 12-point
- ▶ Submit Dashboard and dataset separately on TISSOL.
- ▶ Sourced Figures, Text, Charts, etc. should be properly cited and referenced. Use any standard style (like APA) for referencing.
- ▶ Referencing: Use a standard style (APA / Harvard). All figures, tables, and datasets must be properly cited.
- ▶ Plagiarism will not be tolerated. Direct use of AI-generated text will lead to supplementary assessment.
- ▶ You may use AI tools only for idea organisation or for reference, not for writing content. The submission must reflect your own analysis and language.